United States Patent Application Publication	20250283732
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Chen; Gui et al.

MAP QUALITY ASSESSMENT SYSTEM

Abstract

A map quality assessment system that evaluates error associated with primary map data includes one or more central computers that execute instructions to determine the error associated with the primary map data, compare the error associated with the primary map data with a range of values defined by one or more quality metric values, and in response to determining the error associated with the primary map data falls within the range of values defined by the one or more quality metric values, retain a template for selecting the primary map data. In response to determining the error associated with the primary map data falls outside the range defined by the one or more quality metric values, the one or more central computers evaluate the primary map data for real-life anomalies within one or more roadways represented by the primary map data.

Inventors: **Chen; Gui (Sterling Heights, MI), Bai; Fan (Ann Arbor, MI), Gemar; Mason David (Cedar Park, TX), Hwang; Joon (Pflugerville, TX), Chen; Shu (Rochester Hills, MI), Onifade; Ibrahim (Katy, TX), Ghorban Zadeh; Leila (Seattle, WA), Haller; Christian (Austin, TX)**

Applicant: **GM Global Technology Operations LLC** (Detroit, MI)

Family ID: **96810498**

Appl. No.: **18/599627**

Filed: **March 08, 2024**

Publication Classification

Int. Cl.: **G01C21/00** (20060101)

U.S. Cl.:

CPC **G01C21/387** (20200801); **G01C21/3815** (20200801);

Background/Summary

INTRODUCTION

[0001] The present disclosure relates to a map quality assessment system that evaluates error associated with map data. The map quality assessment system evaluates the error associated with the map data based on gold source map data or, in the alternative, based on a self-evaluation without the gold source map data.

[0002] An autonomous vehicle executes various tasks such as, but not limited to, perception, localization, mapping, path planning, decision making, and motion control. Autonomous vehicles rely upon map data for many of the tasks that are executed such as localization, mapping, and path planning. It is to be appreciated that different versions of map data representing the same geographical area may be generated, where each version of the map data is generated from different data sources.

[0003] One example of a version of map data is based on telemetry data. The telemetry data may be collected from numerous vehicles and combined based on various aggregation algorithms to determine various types of map content such as, for example, global positioning system (GPS) trajectories, and subsequently inferred lane geometries. However, it is to be appreciated that error such as, for example, lateral bias error, random noise, and the like, may exist when the map content is compared to content generated based on ground truth data. The error may cause the map data to indicate an inaccurate position of the lane lines, which in turn may cause issues with localization and may lead to inaccurate decisions that are made by the autonomous vehicle's control system.

[0004] Thus, while maps for autonomous vehicles achieve their intended purpose, there is a need in the art for an improved approach for evaluating map quality.

SUMMARY

[0005] According to several aspects, a map quality assessment system that evaluates error associated with primary map data is disclosed. The map quality assessment system includes one or more central computers in wireless communication one or more communication networks for receiving the primary map data. The one or more central computers executes instructions to determine the error associated with the primary map data, wherein the primary map data represents a predefined geofenced area. The one or more central computers compare the error associated with the primary map data with a range of values defined by one or more quality metric values. In response to determining the error associated with the primary map data falls within the range of values defined by the one or more quality metric values, The one or more central computers retain a template for selecting the primary map data representing the predefined geofenced area. In response to determining the error associated with the primary map data falls outside the range defined by the one or more quality metric values, the one or more central computers evaluate the primary map data for real-life anomalies within one or more roadways represented by the primary map data. In response to determining no real-life anomalies exist within the one or more roadways represented by the primary map data, the one or more central computers implement an updated template for selecting primary map data points representing the predefined geofenced area based on the one or more quality metric values.

[0006] In another aspect, the one or more central computers execute instructions to implement the updated template implemented by: updating weights assigned to the primary map data points of the primary map data based on a corresponding level of accuracy.

[0007] In yet another aspect, the weights corresponding to the primary map data points including a higher level of accuracy based on the one or more quality metric values are assigned a higher weighted value and the weights corresponding to the primary map data points including a lower level of accuracy based on the one or more quality metric values are assigned a lower weighted

value.
[0008] In an aspect, the one or more central computers execute instructions to: determine the error associated with the primary map data based on an absolute offset between primary map data points corresponding to the primary map data and gold source map data points corresponding to gold source map data.
[0009] In another aspect, the one or more central computers execute instructions to: receive road network data representing a road network for the predefined geofenced area, where the road network is a network graph that models roadways based on a plurality of road segments.
[0010] In yet another aspect, the one or more central computers determine the absolute offset between the primary map data points and the gold source map data points based on a bounding box approach that creates a plurality of bounding boxes that each enclose one of the plurality of road segments that are part of the road network.
[0011] In an aspect, the one or more central computers execute instructions to: align the primary map data points and the gold source map data points with one another, where the primary map data points and the gold source map data points are located at cross-sectional ends of the each of the plurality of road segments, and determine the absolute offset between the primary map data points and the gold source map data points for all of the cross-sectional ends of the road segments located within the predefined geofenced area.
[0012] In another aspect, a distance is measured between the cross-sectional ends of the plurality of road segments, and the distance between the cross-sectional ends of each road segment is dimensioned based on a target accuracy of the primary map data.
[0013] In yet another aspect, the one or more central computers execute instructions to: determine a temporal offset between a first set of lane lines and a second set of lane lines, where the temporal offset represents the error associated with the primary map data and is a perpendicular distance measured at the same primary map data point between a first timestep and a second timestep.
[0014] In an aspect, the first set of lane lines are determined at the first timestep and are drawn based on the primary map data, and the second set of lane lines are determined at the second timestep, where the second timestep occurs after the first timestep.
[0015] In another aspect, the one or more central computers execute instructions to: determine a spatial offset between a first set of lane lines and a second set of lane lines, where the spatial offset is measured between an end portion of the first set of lane lines and a beginning position of the second set of lane lines, and the spatial offset represents the error associated with the primary map data.
[0016] In yet another aspect, the first set of lane lines are determined at a first location and the second set of lane lines are determined at a second location positioned directly adjacent to the first location.
[0017] In an aspect, the one or more central computers execute instructions to: determine a likelihood of user intervention that represents a number of times user intervention is required during autonomous driving when compared to a total number of passes by autonomous vehicles for a specific road segment located within the predefined geofenced area, where the likelihood of user intervention is representative of the error associated with the primary map data.
[0018] In another aspect, the likelihood of user intervention is expressed as:
[00001] $P_i = \frac{U_i}{N_i}$
where P.sub.i represents the likelihood of user intervention, U.sub.i represents the number of times user intervention is required, and N.sub.i represents the total number of passes by autonomous vehicles for the specific road segment.
[0019] In yet another aspect, a method for evaluating error associated with primary map data. The method includes determining, by one or more central computers, the error associated with the primary map data, where the primary map data represents a predefined geofenced area and the one or more central computers are in wireless communication one or more communication networks for

receiving the primary map data. The method includes comparing, by the one or more central computers, the error associated with the primary map data with a range of values defined by one or more quality metric values. In response to determining the error associated with the primary map data falls within the range of values defined by the one or more quality metric values, the method includes retaining, by the one or more central computers, a template for selecting the primary map data representing the predefined geofenced area. In response to determining the error associated with the primary map data falls outside the range defined by the one or more quality metric values, the method includes evaluating, by the one or more central computers, the primary map data for real-life anomalies within one or more roadways represented by the primary map data. In response to determining no real-life anomalies exist within the one or more roadways represented by the primary map data, the method includes implementing an updated template for selecting primary map data points representing the predefined geofenced area based on the one or more quality metric values.

[0020] In an aspect, the method further comprises implementing the updated template implemented by: updating weights assigned to the primary map data points of the primary map data based on a corresponding level of accuracy.

[0021] In another aspect, method further includes determining the error associated with the primary map data based on an absolute offset between primary map data points corresponding to the primary map data and gold source map data points corresponding to gold source map data.

[0022] In yet another aspect, the method further includes receiving road network data representing a road network for the predefined geofenced area, wherein the road network is a network graph that models roadways based on a plurality of road segments.

[0023] In an aspect, the method further includes determining the absolute offset between the primary map data points and the gold source map data points based on a bounding box approach that creates a plurality of bounding boxes that each enclose one of the plurality of road segments that are part of the road network.

[0024] In another aspect, the method further includes aligning the primary map data points and the gold source map data points with one another, where the primary map data points and the gold source map data points are located at cross-sectional ends of the each of the plurality of road segments, and determining the absolute offset between the primary map data points and the gold source map data points for all of the cross-sectional ends of the road segments located within the predefined geofenced area.

[0025] Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

[0027] FIG. **1** is a schematic diagram of the disclosed map quality assessment system that includes one or more central computers that receive map data over one or more communication networks, according to an exemplary embodiment;

[0028] FIG. **2** is a block diagram illustrating the software architecture for the one or more central computers shown in FIG. **1**, according to an exemplary embodiment;

[0029] FIG. **3** is schematic diagram of a bounding box determined by the one or more central computers, according to an exemplary embodiment;

[0030] FIG. **4** is a schematic diagram of a road network data including a plurality of cross-

sectioned road segments, according to an exemplary embodiment;
[0031] FIG. **5**A is a schematic diagram illustrating an approach to determine error associated with primary map data based on temporal inconsistencies, according to an exemplary embodiment;
[0032] FIG. **5**B is a schematic diagram illustrating an approach to determine the error associated with the primary map data based on spatial inconsistencies, according to an exemplary embodiment; and
[0033] FIG. **6** is a process flow diagram illustrating a method for evaluating the error associated with the primary map data, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0034] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.

[0035] Referring to FIG. **1**, an exemplary map quality assessment system **10** for evaluating error associated with map data is illustrated. The map quality assessment system **10** includes one or more central computers **20** located at a back-end office **22**, where the one or more central computers **20** are in wireless communication with one or more communication networks **24**. The one or more central computers **20** obtain primary map data via the one or more communication networks **24**. In one embodiment, the one or more central computers **20** may also obtain gold source map data as well via the one or more communication networks **24**. However, it is to be appreciated that in some embodiments the gold source map data may be unavailable.

[0036] The primary map data and the gold source map data both represent the same predefined geofenced area. The predefined geofenced area represents a real-world geographical area that is defined by a virtual perimeter. It is to be appreciated that the primary map data is based on one or more unique sources of data such as, for example, global positioning system (GPS) data, image data collected by an on-board camera of a vehicle, a vehicle telemetry source, aerial or satellite image data, or data collected from survey vehicles. Some examples of versions of primary map data include, but are not limited to, crowdsourced map data, telemetry-based map data, and aerial map data. The gold source map data represents ground truth data or, alternatively, the most accurate and up-to-date version of map data representing the predefined geofenced area currently available. In one non-limiting embodiment, the gold source map data is high-definition map data.

[0037] In the non-limiting embodiment as shown in FIG. **1**, the one or more central computers **20** are in wireless communication with one or more autonomous vehicles **26**. Although autonomous vehicles are described, it is to be appreciated that semi-autonomous vehicles that are equipped with an advanced driver assistance system (ADAS) may be included as well. After evaluating the primary map data, the one or more central computers **20** may share a vehicle map that is generated based on the primary map data with the one or more autonomous vehicles **26**. As explained below, the one or more central computers **20** evaluate the error associated with the primary map data based on an absolute offset between primary map data points and gold source map data points. Alternatively, in another embodiment the one or more central computers **20** determine the error associated with the primary map data based on self-evaluation without the gold source map data. In yet another embodiment, the error associated with the primary map data is based on a likelihood of user intervention during autonomous driving. It is to be appreciated that the error associated with the primary map data may represent any type of error that is associated with map data such as, but not limited to, lateral bias error or perception error. In one embodiment, the primary map data is created specifically for an autonomous driving system such as, for example, an automated driving system (ADS) or ADAS.

[0038] FIG. **2** is a block diagram illustrating the software architecture of the one or more central computers **20** shown in FIG. **1**. In the example as shown in FIG. **2**, the one or more central computers **20** includes a gold source evaluation module **30**, a self-evaluation module **32**, and a selection module **34**. As explained below, the gold source evaluation module **30** compares the primary map data with the gold source map data to determine the error associated with the primary map data, while the self-evaluation module **32** determines the error when the gold source map data is unavailable. Referring to both FIGS. **1** and **2**, the gold source evaluation module **30** of the one or more central computers **20** includes a bounding box evaluation submodule **40** that compares the primary map data with the gold source map data based on a bounding box approach and a cross-sectional evaluation submodule **42** that compares the primary map data with the gold source map data based on a cross-sectioned approach, which are both described below.

[0039] The one or more central computers **20** receives road network data representing a road network of the predefined geofenced area as input from the one or more communication networks **24**, where the road network is a network graph that models roadways based on a plurality of road segments. The one or more central computers **20** also receives the primary map data and the gold source map data that both represent the predefined geofenced area from the one or more communication networks **24** as well. One example of road network data is the open street map (OSM), however, it is to be appreciated that other types of road network data may be used as well.

[0040] Evaluating the primary map data for the absolute offset between the primary map data with the gold source map data based on a bounding box approach shall now be described. The bounding box evaluation submodule **40** of the one or more central computers **20** divides the road network data into a plurality of road segments **52**, which is illustrated in FIG. **3**. Referring to both FIGS. **2** and **3**, each of the plurality of road segments **52** include the same length, where the length of each road segment **52** may be equal to an average accurate perception range of a vehicle **26**. In one non-limiting embodiment, the length is about fifty meters. The bounding box evaluation submodule **40** of the one or more central computers **20** creates bounding boxes **64** corresponding to the primary map data and the gold source map data. Each bounding box **64** has a width **48** and a height **50** (i.e., a maximum height). Each bounding box **64** encompasses one road segment **52**, and the width **48** of each bounding box **64** is equal to the length of the road segment **52**. The height **50** of the bounding box **64** may be greater than the width **48** of the bounding box **64**.

[0041] With reference to FIG. **3**, the bounding boxes **64** encompass the road segment **52**. Each road segment **52** includes at least two road-segment points (i.e., the first road-segment point **66** and the second road-segment point **68**) that define the boundaries of the road segment **52**. In FIG. **3**, the bounding box **64** includes a first linear boundary **70**, a second linear boundary **72**, a third linear boundary **74**, and a fourth linear boundary **76**. The first linear boundary **70** intersects the first road-segment point **66**. The second linear boundary **72** intersects the second road-segment point **68**. The first linear boundary **70** is parallel to the second linear boundary **72**. The third linear boundary **74** is parallel to the fourth linear boundary **76**. The distance **78** from the third linear boundary to the first road-segment point **66** along the first direction y′ is equal to the distance **80** from the second road-segment point **68** to the fourth linear boundary **76** along the first direction y′. The height **50** (i.e., maximum height) of the bounding box extends from the third linear boundary **74** to the fourth linear boundary **76**. The distance **80** from the second road-segment point **68** to the fourth linear boundary **76** along the first direction y′ is half of the maximum height of the first bounding box. The road segment **52** is parallel to a second direction x′. The second direction x′ is orthogonal to the first direction y′ the first direction y′ and the second direction x′ may be defined as axes that form a vehicle coordinate system **82** for the bounding box **64**.

[0042] As discussed above, the first road-segment point **66** and the second road-segment point **68** define the extreme ends (i.e., termini) of the road segment **52**. The bounding box evaluation submodule **40** of the one or more central computers **20** uses the first road-segment point **66** and the second road-segment point **68** to create the bounding box **64**. In the vehicle coordinate system **82**, the x′-axis (which is defined by the x′ direction) is parallel to the road segment **52**, and the y′-axis (which is defined by the y′ direction is perpendicular to the road segment **52**. The angle θ is the heading of a vehicle **26**. In vehicle coordinate system **82**, the first road-segment point **66** is at coordinate (x1, y1) of the vehicle coordinate system **82**, and the second road-segment point **68** is at coordinate (x2, y2). The distance **80** from the second road-segment point **68** to the fourth linear boundary **76** along the first direction y′ is half of the height **50** of the bounding box **64**. Distance **80** from the second road-segment point **68** to the fourth linear boundary **76** is constant for all the bounding boxes **64**. Further, the distance **80** from the second road-segment point **68** to the fourth linear boundary **76** along the first direction y′ may be represented by the letter "d". The bounding box defines four corner points (i.e., first corner point Q**1** at coordinate (x1, y1+d), second corner point Q**2** at coordinate (x2, y2+d), third corner point Q**3** (x1, y1−d), and fourth corner point Q**4** (x2, y1−d).

[0043] With continued reference to FIG. **3**, the coordinate of the origin of the vehicle coordinate system **82** is (m, n). The origin of the vehicle coordinate system **82** is translated to the origin of a global coordinate system **84**. At this juncture, the coordinates of the bounding boxes will be the following: first corner point Q**1** (x1+m, y1+d+n), second corner point Q**2** (x2+m, y2+d+n), third corner point Q**3** (x1+m, y1−d+n), and fourth corner point Q**4** (x2+m, y2−d+n). Then, the vehicle coordinate system **82** is rotated such that the x′-axis of the vehicle coordinate system **82** aligns with the x-axis of the global coordinate system **84** and the y′-axis of the vehicle coordinate system **82** aligns of the y-axis of the global coordinate system **84**. In this rotation, the rotation angle in clockwise is e (which is the heading of the vehicle **26**). After the rotation, in the global coordinate system xy, the coordinates of the bounding box **64** are: the first corner point Q**1** ((x1+m)*cos θ+(y1+d+n)*sin θ, (y1+d+n)*cos θ−(x1+m)*sin θ), the second corner point Q**2** ((x2+m)*cos θ+(y2+d+n)*sin θ, (y2+d+n)*cos θ−(x2+m)*sin θ), the third corner point Q**3** ((x1+m)*cos θ+(y1−d+n)*sin θ, (y1−d+n)*cos θ−(x1+m)*sin θ), and the fourth corner point Q**4** ((x2+m)*cos θ+(y2−d+n)*sin θ, (y2−d+n)*cos θ−(x2+m)*sin θ).

[0044] The bounding box evaluation submodule **40** of the one or more central computers **20** creates a primary map tile and a gold source map tile based on the primary map data and the gold source map data, respectively, by filtering out the map data inside the bounding box **64**. The primary map tile is obtained by filtering the primary map data, and the gold source map tile is obtained by filtering the gold source map data. For all the bounding boxes **64**, the bounding box evaluation submodule **40** of the one or more central computers **20** creates the primary map tile and the gold source map tile from the primary map data and the gold source map data, respectively. The filtered data to create the map tiles (i.e., the primary map tile and the gold source map tile) may be part of the map data inside of the bounding box **64** but outside the lane lines.

[0045] The bounding box evaluation submodule **40** of the one or more central computers **20** executes a point cloud registration to align the plurality of primary map data points in the primary map tile with the plurality of gold source map data points in the gold source map tile using rotation and translation transformations to determine the absolute offsets between the primary map data points of the primary map data and the gold source map data points of the gold source map data. Specifically, an iterative closest point (ICP) process or algorithm may align the plurality of primary map data points in the primary map tile with the plurality of gold source map data points in the gold source map tile using rotation and translation transformations. Furthermore, a random initialization process may be used to align the primary map data points with the gold source map data points based on the point cloud registration.

[0046] The bounding box evaluation submodule **40** of the one or more central computers **20** builds a KD tree with the primary map data points using a KD algorithm, where the primary map points data are created first as the line segments. Then, the center points of the line segments are obtained and designated as the KD tree nodes. For each point for the gold source map data, the nearest point by KD tree query and the corresponding line segments from the primary map data is determined. Then, the offset from this point on the gold source map data to the line segments from the first map is calculated. The bounding box evaluation submodule **40** of the one or more central computers **20** determines, based on the KD tree, the absolute offsets from each of the plurality of primary map data points to each corresponding gold source map data points. To do so, the bounding box evaluation submodule **40** of the one or more central computers **20** calculates the distance from one of the primary map data points of the primary map data to the corresponding gold source map data point of the gold source map data. This is repeated from all to determine all absolute offsets between the primary map data points and the gold source map data points. The bounding box evaluation submodule **40** of the one or more central computers **20** then determines the relative map error based on a histogram of all of the absolute offsets.

[0047] An approach to determine the absolute offset between the primary map data points and the gold source map data points based on a cross-sectioned approach shall now be described. FIG. **4** is an illustration of the road network data of the predefined geofenced area, where the roadways located within the predefined geofenced area are represented by a plurality of road segments **92**, a plurality of nodes **94**, and opposing road edges **98**. The plurality of nodes **94** each represent a cross-sectional end **96** of one of the road segments **92**. The opposing road edges **98** represent theoretical road edges, and not the opposing topological graph edges. It is to be appreciated that the opposing road edges **98** include a unique edge identifier (ID), where each edge ID identifies a pair of opposing road edges **98** corresponding to a particular road segment **92**. A distance D is measured between the cross-sectional ends **96** of the road segments **92**, where the distance D between the cross-sectional ends **96** of each road segment **92** is dimensioned based on a target accuracy of the primary map data. Merely by way of example, if the primary map data is created for an autonomous driving system such as ADS or ADAS, then the distance D between the cross-sectional ends **96** of the road segments **92** is about one meter.

[0048] The cross-sectional evaluation submodule **42** of the one or more central computers **20** determines the absolute offset between primary map data points **100** that correspond to the primary map data and gold source map data points **102** that correspond to the gold source map data by first aligning the primary map data points **100** and the gold source map data points **102** with one another. As seen in FIG. **4**, the primary map data points **100** and the gold source map data points **102** are located at the cross-sectional ends **96** of the road segments **92**. The cross-sectional evaluation submodule of the one or more central computers **20** aligns the primary map data points **100** and the gold source map data points **102** with one another by executing one or more map matching algorithms to perform topology. As seen in FIG. **4**, the cross-sectional ends **96** of each road segment **92** intersect one or more primary map data points **100** and one or more corresponding gold source map data points **102**. The cross-sectional evaluation submodule **42** of the one or more central computers **20** then determines an absolute offset between the one or more primary map data points **100** and the one or more corresponding gold source map data points **102** for all of the cross-sectional ends **96** of the road segments **92** located within the predefined geofenced area. The cross-sectional evaluation submodule **42** of the one or more central computers **20** then determines a sum of all the absolute offsets for each road segment **92** within the predefined geofenced area and divides the sum of all the absolute offsets with a total number of road segment **92** within the predefined geofenced area to determine the relative error.

[0049] As mentioned above, the self-evaluation module **32** determines map error when the gold source map data is unavailable. Referring to both FIGS. **1** and **2**, the self-evaluation module **32** of the one or more central computers **20** includes a temporal evaluation submodule **44** that determines the error of the primary map data based on temporal inconsistencies, a spatial evaluation submodule **46** that determines the error of the primary map data based on spatial inconsistencies, and a user intervention submodule **60** that determines the error of the primary map data based user intervention during autonomous or semi-autonomous driving. It is to be appreciated that while FIG. **2** illustrates the one or more central computers **20** determining the error based on temporal inconsistencies or spatial inconsistencies, the error may be determined locally by one or more controllers that are part of one of the autonomous vehicles **26** (FIG. **1**) instead.

[0050] Determining the error based on temporal inconsistencies shall now be described. FIG. **5**A is a schematic diagram of a first set of lane lines **110** determined at a first timestep T that are drawn based on the primary map data. FIG. **5**A also includes a second set of lane lines **112** determined at a second timestep T+1, where the second timestep T+1 occurs after the first timestep T. Both the first set of lane lines **110** and the second set of lane lines **112** represent the same roadway.

[0051] Referring to both FIGS. **2** and **5**A, the temporal evaluation submodule **44** of the one or more central computers **20** determines a temporal offset **114** between the first set of lane lines **110** and the second set of lane lines **112**, where the temporal offset **114** represents the error of the primary map data. The temporal offset **114** represents a perpendicular distance that is measured at the same primary map data point between the first timestep T and the second timestep T+1. The temporal evaluation submodule **44** of the one or more central computers **20** then compares the temporal offset **114** with a temporal threshold value, where the temporal threshold value is selected based on a target level of accuracy. It is to be appreciated that the target level of accuracy is determined based on the specific application of the primary map data. In one embodiment, if the temporal error associated with any of the primary map data points exceeds the temporal threshold value, then a notification is generated.

[0052] In one embodiment, the temporal evaluation submodule **44** of the one or more central computers **20** selects the maximum temporal offset value from the primary map data points as the overall temporal offset of the primary map data. Specifically, Equation 1 determines the overall temporal offset of the primary map data, which is as follows:

[00002] $$\Pi_T = \max_i \text{.Math.} \overset{\text{.Math.}}{A_i} - \frac{\overset{\text{.Math.}}{B_i} \text{.Math.} \overset{\text{.Math.}}{A_i}}{\text{.Math.} \overset{\text{.Math.}}{A}_i \text{.Math.}} \text{.Math.} \quad \text{Equation1}$$

where II.sub.T represents the overall temporal offset of the primary map data, i represents the total number of primary map data points that define the lane lines **110**, **112**, custom-character represents a point along the the first set of lane lines **110** at the first timestep T, and custom-character represents a point along the second set of lane lines **112** at the second timestep T**2**.

[0053] Determining the error based on spatial inconsistencies shall now be described. FIG. **5**B is a schematic diagram of a first set of lane lines **120** determined at a first location S that are drawn based on the primary map data. FIG. **5**B also includes a second set of lane lines **122** based on the primary map data that are determined at a second location S+1, where the second location S+1 is positioned directly adjacent to the first location S. It is to be appreciated that the second location S+1 is positioned directly adjacent to the first location S in either the lateral or the longitudinal direction.

[0054] Referring to FIGS. **2** and **5**B, the spatial evaluation submodule **46** of the one or more central computers **20** determines a spatial offset **124** between the first set of lane lines **120** and the second set of lane lines **122**, where the spatial offset **124** is measured between an end portion **126** of the first set of lane lines **120** and a beginning position **128** of the second set of lane lines **122** where the first set and the second set of lane lines **120**, **122** overlap. The spatial offset **124** represents the error associated with the primary map data. The spatial evaluation submodule **46** of the one or more central computers **20** then compares the spatial offset **124** with a spatial threshold value, where the spatial threshold value is selected based on the target level of accuracy described above. In one embodiment, if the spatial error associated with any of the primary map data points exceeds the spatial threshold value, then a notification is generated.

[0055] In one embodiment, the spatial evaluation submodule **46** of the one or more central computers **20** determines the spatial offset **124** based on Equation 2, which is as follows:

[00003] $$\Pi_S = \text{.Math.} \overset{\text{.Math.}}{A}_{End} - \frac{\overset{\text{.Math.}}{B}_{\text{Beginning}} \text{.Math.} \overset{\text{.Math.}}{A}_{End}}{\text{.Math.} \overset{\text{.Math.}}{A}_{End} \text{.Math.}} \text{.Math.} \quad \text{Equation2}$$

where II.sub.s represents the overall spatial offset of the primary data points, custom-character represents a point at the end portion **126** of the first set of lane lines **120**, and custom-character represents a point at the beginning position **128** of the second set of lane lines **122**.

[0056] Referring to FIG. **2**, the user intervention submodule **60** of the one or more central computers **20** determines a likelihood of user intervention during autonomous driving, where the likelihood of user intervention is representative of the error associated with the primary map data. This is because when the error associated with the primary map data is relatively low, the autonomous vehicle **26** (FIG. **1**) may be navigated with minimal or no user intervention. However, as the error increases, the instances when a user intervenes by performing one or more manual driving maneuvers to navigate the vehicle during autonomous driving increases. The user intervention represents one or more manual driving maneuvers that are executed by a user to accommodate the error associated with the primary map data. Some examples of map error that may cause the user to execute one or more manual driving maneuvers include, but are not limited to, map curvature error, lane line inaccuracies, perception issues, missing signage, and the wrong location of signage.

[0057] In one embodiment, the likelihood of user intervention represents a number of times user intervention is required during autonomous driving when compared to a total number of passes by autonomous vehicles for a specific road segment located within the predefined geofenced area. Specifically, in one embodiment, the likelihood of user intervention is expressed in Equation 3, and is as follows:

[00004] $P_i = \frac{U_i}{N_i}$ Equation3

where P.sub.i represents the likelihood of user intervention, U.sub.i represents the number of times user intervention is required, and N.sub.i represents the total number of passes by autonomous vehicles for the specific road segment.

[0058] Referring to FIG. **2**, the selection module **34** of the one or more central computers **20** receives the error associated with the primary map data from either the gold source evaluation module **30** or the self-evaluation module **32** and compares the error associated with the primary map data with one or more quality metric values. The quality metric values are each based on a specific quality metric that is indicative of the quality of the primary map data and are based on the target level of accuracy of the primary map data. As mentioned above, the target level of accuracy is determined based on the specific application of the primary map data. Merely by way of example, in one embodiment the quality metric values indicate an estimated horizontal position error (EHPE) of the primary map data, however, it is to be appreciated that other measures of quality may be used as well such as, for example, confidence values. In one non-limiting embodiment, the quality metric values may be defined by a range of values. For example, the range of quality metric values may be expressed as a number ranging from 0 to 9, where 0 represents the greatest level of accuracy and 9 represents the lowest level of accuracy. In one embodiment, the one or more quality metric values may be weighted based on importance, where primary map data points including a higher level of accuracy are assigned a higher importance when compared to primary map data points including a lower level of accuracy.

[0059] The selection module **34** of the one or more central computers **20** compares the error associated with the primary map data with the one or more quality metric values. In response to determining the error associated with the primary map data falls within the range of values defined by the one or more quality metric values, the selection module **34** retains a template for selecting the primary map data representing the predefined geofenced area based on the one or more quality metric values. The template for selecting the primary map data points assigns weights to the primary map data points, where greater weight is assigned to the primary map data points including a higher level of accuracy and less weight is assigned to the primary map data points that include a lower level of accuracy. The selection module **34** of the one or more central computers **20** may then generate a vehicle map based on the primary map data. In embodiments, the vehicle map may be shared with the one or more autonomous vehicles **26** (FIG. **1**).

[0060] In response to determining the error associated with the primary map data that falls outside the range defined by the one or more quality metric values, the selection module **34** of the one or more central computers **20** then evaluates the primary map data for real-life anomalies within one or more roadways represented by the primary map data. The real-life anomalies within the one or more roadways may adversely affect the accuracy of the primary map data. Some examples of the real-life anomalies within the one or more roadways include, but are not limited to, faded lane lanes, repainted lane lines, construction activity, road closures, and traffic incidents. In response to detecting one or more real-life anomalies within the one or more roadways represented by the primary map data, the selection module **34** of the one or more central computers **20** continues to monitor the primary map data until the real-life anomalies no longer exist, and then collects a new set of primary map data for evaluation.

[0061] In response to determining no real-life anomalies exist within the one or more roadways represented by the primary map data, the selection module **34** of the one or more central computers **20** implements an updated template for selecting the primary map data points representing the predefined geofenced area based on the one or more quality metric values. The updated template is implemented by updating the weights assigned to the primary map data points of the primary map data based on a corresponding level of accuracy. Specifically, the weights corresponding to the primary map data points including a higher level of accuracy based on the one or more quality metric values are assigned a higher weighted value, while the weights corresponding to the primary map data points including a lower level of accuracy based on the one or more quality metric values are assigned a lower weighted value.

[0062] In one embodiment, the weights are assigned to the primary map data points based on a probability distribution to ensure that the primary map data points include a mix of different levels of error. In an embodiment, the updated template is determined based on an iterative process, where a selection criteria to determine the new weighted values is refined during each iteration until the error associated with the primary map data falls within the range defined by the one or more quality metric values.

[0063] FIG. **6** is a process flow diagram illustrating a method **600** for evaluating the error associated with the primary map data by the map quality assessment system **10** shown in FIG. **1**. Referring generally to FIGS. **1-6**, the method **600** may begin at block **602**. In block **602**, the one or more central computers **20** determine the error associated with the primary map data, where the primary map data represents the predefined geofenced area. As mentioned above, the one or more central computers **20** may determine the error associated with the primary map data based on a variety of different approaches. Specifically, in one embodiment, the error is an absolute offset between the primary map data points and the gold source map data points. In another embodiment, the one or more central computers **20** determine the error associated with the primary map data based on self-evaluation without the gold source map data. In yet another embodiment, the error associated with the primary map data is based on a likelihood of user intervention during autonomous driving. The method **600** may then proceed to block **604**.

[0064] In block **604**, the the selection module **34** of the one or more central computers **20** compares the error associated with the primary map data with the range of values defined by the one or more quality metric values. The method **600** may then proceed to decision block **606**.

[0065] In decision block **606**, in response to determining the error associated with the primary map data falls within the range of values defined by the one or more quality metric values, the method **600** proceeds to block **608**. In block **608**, the selection module **34** of the one or more central computers **20** retains the template for selecting the primary map data representing the predefined geofenced area. In an embodiment, the one or more central computers **20** may generate a vehicle map based on the primary map data that is transmitted to the autonomous vehicle **26** located within the predefined geofenced area. The method **600** may then terminate.

[0066] Returning back to decision block **606**, in response to determining the error associated with the primary map data falls outside the range defined by the one or more quality metric values, the method **600** proceeds to block **610**. The block **610**, the selection module **34** of the one or more central computers **20** evaluates the primary map data for real-life anomalies within one or more roadways represented by the primary map data. The method **600** may then proceed to decision block **612**.

[0067] In decision block **612**, in response to detecting one or more real-life anomalies within the one or more roadways represented by the primary map data, the method **600** proceeds to block **614**. In block **614**, the selection module **34** of the one or more central computers **20** continues to monitor the primary map data until the real-life anomalies no longer exist, and then collects a new set of primary map data for evaluation. The method **600** may then return to block **602**.

[0068] Referring back to decision block **612**, in response to determining no real-life anomalies exist within the one or more roadways represented by the primary map data, the method **600** proceeds to block **616**. In block **616**, the selection module **34** of the one or more central computers **20** implements the updated template for selecting the primary map data points representing the predefined geofenced area based on the one or more quality metric values. As mentioned above, the updated template is implemented by updating the weights assigned to the primary map data points of the primary map data based on a corresponding level of accuracy. The method **600** may then terminate.

[0069] Referring generally to the figures, the disclosed map quality assessment system provides various technical effects and benefits. Specifically, the map quality assessment system provides an approach to evaluate the error associated with map data. It is to be appreciated that the map quality assessment system provides a variety of approaches to determine the error associated with the map data. In particular, the error may be determined based on gold source map data or, in the alternative, based on a self-evaluation when the gold source map data is unavailable.

[0070] The central computers may refer to, or be part of an electronic circuit, a combinational logic circuit, a field programmable gate array (FPGA), a processor (shared, dedicated, or group) that executes code, or a combination of some or all of the above, such as in a system-on-chip. Additionally, the controllers may be microprocessor-based such as a computer having a at least one processor, memory (RAM and/or ROM), and associated input and output buses. The processor may operate under the control of an operating system that resides in memory. The operating system may manage computer resources so that computer program code embodied as one or more computer software applications, such as an application residing in memory, may have instructions executed by the processor. In an alternative embodiment, the processor may execute the application directly, in which case the operating system may be omitted.

[0071] The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. A map quality assessment system that evaluates error associated with primary map data, the map quality assessment system comprising: one or more central computers in wireless communication one or more communication networks for receiving the primary map data, the one or more central computers executing instructions to: determine the error associated with the primary map data, wherein the primary map data represents a predefined geofenced area; compare the error associated with the primary map data with a range of values defined by one or more quality metric values; in response to determining the error associated with the primary map data falls within the range of values defined by the one or more quality metric values, retain a template for selecting the primary map data representing the predefined geofenced area; in response to determining the error associated with the primary map data falls outside the range defined by the one or more quality metric values, evaluate the primary map data for real-life anomalies within one or more roadways represented by the primary map data; and in response to determining no real-life anomalies exist within the one or more roadways represented by the primary map data, implement an updated template for selecting primary map data points representing the predefined geofenced area based on the one or more quality metric values.

2. The map quality assessment system of claim 1, wherein the one or more central computers execute instructions to implement the updated template implemented by: updating weights assigned to the primary map data points of the primary map data based on a corresponding level of accuracy.

3. The map quality assessment system of claim 2, wherein the weights corresponding to the primary map data points including a higher level of accuracy based on the one or more quality metric values are assigned a higher weighted value and the weights corresponding to the primary map data points including a lower level of accuracy based on the one or more quality metric values are assigned a lower weighted value.

4. The map quality assessment system of claim 1, wherein the one or more central computers execute instructions to: determine the error associated with the primary map data based on an absolute offset between primary map data points corresponding to the primary map data and gold source map data points corresponding to gold source map data.

5. The map quality assessment system of claim 4, wherein the one or more central computers execute instructions to: receive road network data representing a road network for the predefined geofenced area, wherein the road network is a network graph that models roadways based on a plurality of road segments.

6. The map quality assessment system of claim 5, wherein the one or more central computers determine the absolute offset between the primary map data points and the gold source map data points based on a bounding box approach that creates a plurality of bounding boxes that each enclose one of the plurality of road segments that are part of the road network.

7. The map quality assessment system of claim 5, wherein the one or more central computers execute instructions to: align the primary map data points and the gold source map data points with one another, wherein the primary map data points and the gold source map data points are located at cross-sectional ends of the each of the plurality of road segments; and determine the absolute offset between the primary map data points and the gold source map data points for all of the cross-sectional ends of the road segments located within the predefined geofenced area.

8. The map quality assessment system of claim 7, wherein a distance is measured between the cross-sectional ends of the plurality of road segments, and wherein the distance between the cross-sectional ends of each road segment is dimensioned based on a target accuracy of the primary map data.

9. The map quality assessment system of claim 1, wherein the one or more central computers execute instructions to: determine a temporal offset between a first set of lane lines and a second set of lane lines, wherein the temporal offset represents the error associated with the primary map data and is a perpendicular distance measured at the same primary map data point between a first timestep and a second timestep.

10. The map quality assessment system of claim 9, wherein the first set of lane lines are determined at the first timestep and are drawn based on the primary map data, and the second set of lane lines are determined at the second timestep, wherein the second timestep occurs after the first timestep.

11. The map quality assessment system of claim 1, wherein the one or more central computers execute instructions to: determine a spatial offset between a first set of lane lines and a second set of lane lines, wherein the spatial offset is measured between an end portion of the first set of lane lines and a beginning position of the second set of lane lines, and wherein the spatial offset represents the error associated with the primary map data.

12. The map quality assessment system of claim 11, wherein the first set of lane lines are determined at a first location and the second set of lane lines are determined at a second location positioned directly adjacent to the first location.

13. The map quality assessment system of claim 1, wherein the one or more central computers execute instructions to: determine a likelihood of user intervention that represents a number of times user intervention is required during autonomous driving when compared to a total number of passes by autonomous vehicles for a specific road segment located within the predefined geofenced area, wherein the likelihood of user intervention is representative of the error associated with the primary map data.
14. The map quality assessment system of claim 13, wherein the likelihood of user intervention is expressed as: $P_i = \frac{U_i}{N_i}$ wherein P.sub.i represents the likelihood of user intervention, U.sub.i represents the number of times user intervention is required, and N.sub.i represents the total number of passes by autonomous vehicles for the specific road segment.
15. A method for evaluating error associated with primary map data, the method comprising: determining, by one or more central computers, the error associated with the primary map data, wherein the primary map data represents a predefined geofenced area and the one or more central computers are in wireless communication one or more communication networks for receiving the primary map data; comparing, by the one or more central computers, the error associated with the primary map data with a range of values defined by one or more quality metric values; in response to determining the error associated with the primary map data falls within the range of values defined by the one or more quality metric values, retaining, by the one or more central computers, a template for selecting the primary map data representing the predefined geofenced area; in response to determining the error associated with the primary map data falls outside the range defined by the one or more quality metric values, evaluating, by the one or more central computers, the primary map data for real-life anomalies within one or more roadways represented by the primary map data; and in response to determining no real-life anomalies exist within the one or more roadways represented by the primary map data, implementing an updated template for selecting primary map data points representing the predefined geofenced area based on the one or more quality metric values.
16. The method of claim 15, wherein the method further comprises implementing the updated template implemented by: updating weights assigned to the primary map data points of the primary map data based on a corresponding level of accuracy.
17. The method of claim 15, wherein the method further comprises: determining the error associated with the primary map data based on an absolute offset between primary map data points corresponding to the primary map data and gold source map data points corresponding to gold source map data.
18. The method of claim 17, wherein the method further comprises: receiving road network data representing a road network for the predefined geofenced area, wherein the road network is a network graph that models roadways based on a plurality of road segments.
19. The method of claim 18, wherein the method further comprises: determining the absolute offset between the primary map data points and the gold source map data points based on a bounding box approach that creates a plurality of bounding boxes that each enclose one of the plurality of road segments that are part of the road network.
20. The method of claim 18, wherein the method further comprises: aligning the primary map data points and the gold source map data points with one another, wherein the primary map data points and the gold source map data points are located at cross-sectional ends of the each of the plurality of road segments; and determining the absolute offset between the primary map data points and the gold source map data points for all of the cross-sectional ends of the road segments located within the predefined geofenced area.